

Variation in Many-Analyst Projects: Problematic or Helpful? A Multiverse Approach to Address Different Sources of Variation

Thesis Proposal

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1 Introduction

Everything that is based on ‘proper’ science must be true, right? As long as the theory on which it is based is valid and some statistics are applied, the result is assumed to be correct and reproducible. However, recent studies have shown considerable variation in outcomes when multiple researchers answer the same question using the same data, raising concerns about the reproducibility and credibility of science. For example, Silberzahn et al. (2018) asked 29 teams to investigate whether **players** with dark-skin are more likely to receive red cards from the referees than players with light-skin. The results varied widely, with 69% finding a significantly positive effect and 31% not finding any significant effect. Furthermore, in a study where 16 teams were asked to estimate the effect of marital status on cardiovascular outcomes, the results varied from **negative to positive effects** (Kowall et al., 2025).

Many-analyst projects seem to be a good way to gain insight into naturally existing variation in research, as well as the robustness of analyses. However, various studies have failed to identify where exactly this variation stems from. According to Silberzahn et al. (2018), factors such as **prior beliefs**, level of experience or peer-rated quality of the analysis did not seem to explain the variation, while almost all researchers included different covariates. Variation **in** (Kowall et al., 2025) seemed to stem mainly from the analysts’ choice **of longitudinal versus cross-sectional data**, rather than variable definitions or statistical models. Interpreting variation within and across many-analyst projects is difficult – is there a **“hidden universe of uncertainty”** that poses a problem for credible and reproducible science or is the variation meaningful and can it drive the

Table 1: Sources of variation according to Auspurg and Brüderl (2024)

Problematic	Helpful
(a) Model uncertainty including subjective choices in data handling	(c) Substantively meaningful variation, based on different definitions of dependent, independent and confounding variables and/or target populations
(b) Coding Errors	(changes over time)

advancement of theories?

According to Auspurg and Brüderl (2024), it is important to clearly define different sources of variation in many-analyst projects. They distinguish between three sources (table 1). Multiverse analysis is a way to systematically map a wide variety of reasonable ‘research paths’, with the aim of increasing transparency about subjective choices and their impact on the associated results. With this method it is possible to distinguish different sources of variation and evaluate their impact on the results. Hopefully, this will increase awareness of the sensitivity of some statistical paths and the robustness of their estimated results, as well as knowledge about the degree to which different sources of variation play a role in the variation obtained from many-analyst projects (Steegen et al., 2016). Because it is difficult to systematically evaluate code errors, I will focus only on (a) and (c), resulting in the following research question.

Research Question

Based on multiverse analysis, how do different sources of variation, including handling missing data, choice of statistical model and the definition of variables and target population (longitudinal versus cross-sectional), contribute to variation in many-analyst projects?

Expectations

As with previous multiverse and multi-analyst studies, I expect considerable variation depending on the analytical choices. Because the explanations for the variation in previous studies varied widely, and may differ across studies, I have reservations about making explicit expectations in this regard. However, since I will be basing my study on Kowall et al. (2025), I expect to find similar patterns — although this may not be the case, as they only included 16 teams, and the multiverse analysis will be more extensive.

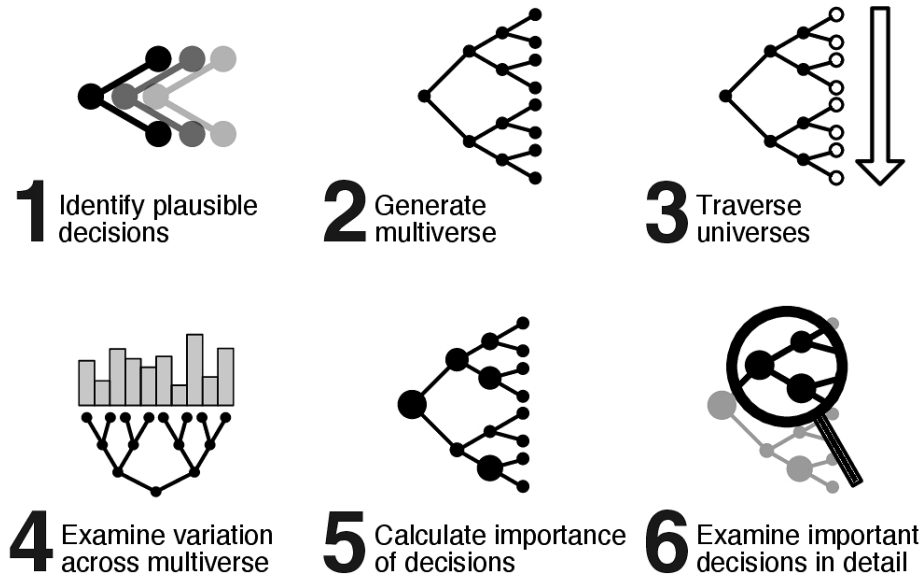


Figure 1: Comprehensive Overview of Multiverse Analysis by Simson et al. (2024)

2 Analytic Strategy

Analysis Plan

Because the study by Kowall et al. (2025) includes many different sources of variation, as researchers used different definitions of their variables, exclusion criteria, methods of missing data handling, statistical models, and type of data (longitudinal versus cross-sectional data), I decided to base my multiverse analysis on this study (see [table 2](#) for the exact specifications).

For a comprehensive overview of the steps involved in multiverse analysis, see figure 1 (Simson et al., 2024).

Data

Like Kowall et al. (2025), I will make use of the SHARE dataset.

Software

All analyses will be performed in RStudio (Team, 2024). I will likely use (at least) the following packages:

- tidyverse (Wickham et al., 2019)
- multiverse (Sarma et al., 2021)

- `spectr` (Masur and Scharkow, 2020)
- `lme4` (Bates et al., 2015)
- `survival` (Therneau, 2024)
- `geepack` (Halekoh et al., 2006)

Ethics

This project is under review by the Faculty Ethics Review Committee (FETC) of the Faculty of Social Sciences, with registration number: XXX .

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